

# Technical Report

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# 1. Introduction

In this deliverable, we applied Creational Design Patterns in our School Management System project.

The goal is to solve different system problems using:

- Singleton Pattern (2 cases)
- Factory Method Pattern (1 case)
- Abstract Factory Pattern (2 cases)

Each case includes:

- Problem description
- Pattern justification
- UML Class Diagram
- Explanation of the solution

## **2. Development Environment**

The project was developed using:

- Java
- NetBeans / Eclipse
- Git
- Maven
- JIRA (for task tracking)

The system represents a School Management System that allows viewing grades, attendance, announcements, and managing user accounts.

## 3. Singleton Pattern

### Case 1 — View Grades

#### Problem

The system must allow students to view grades.

Only one service should access the grades database at a time to avoid conflicts.

#### Solution

We used Singleton pattern to ensure only one instance of GradeViewerService exists.

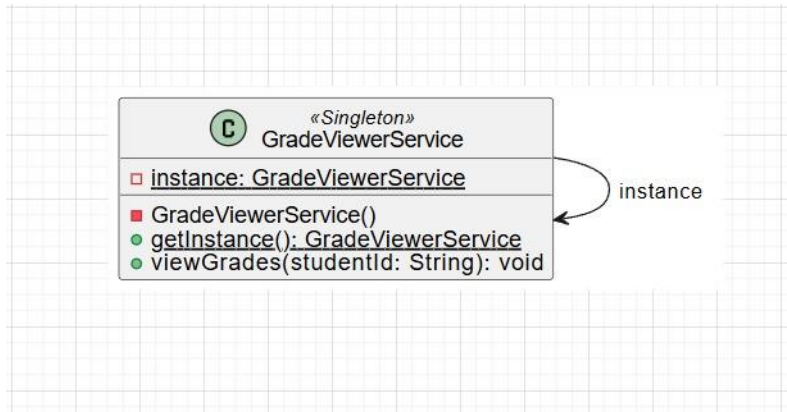
#### Why Singleton?

- Only one DB reader needed
- Prevent multiple connections
- Improve performance

#### Classes

- GradeViewerService

#### UML Diagram



## Explanation

The constructor is private, and the instance is created using `getInstance()`.

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## Case 2 — View Attendance

### Problem

Attendance data must be read safely without creating multiple service objects.

### Solution

Singleton pattern ensures one `AttendanceViewerService`.

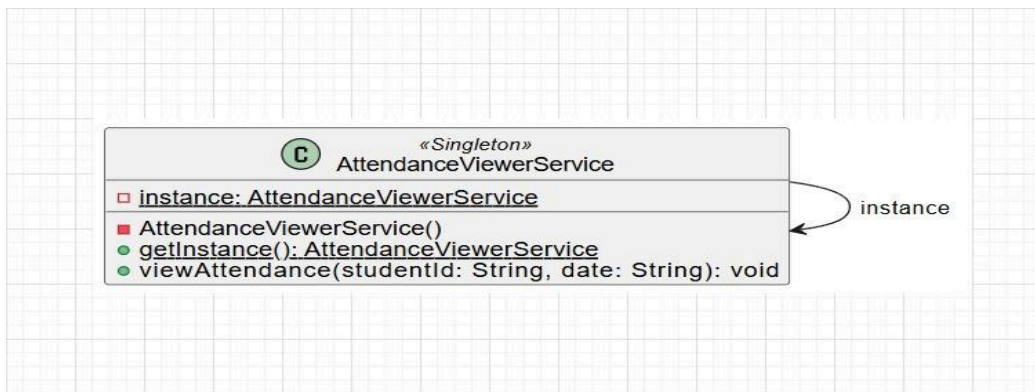
### Why Singleton?

- Avoid multiple DB queries
- Keep service centralized
- Thread-safe access

## Classes

- AttendanceViewerService

## UML Diagram



## Explanation

Double-checked locking is used for thread safety.

# 4. Factory Method Pattern

## Case — Announcements System

### Problem

The system can send announcements in different ways:

- Dashboard
- Email

We don't want client code to know which class to create.

### Solution

Factory Method pattern creates the correct announcement type.

### Why Factory Method?

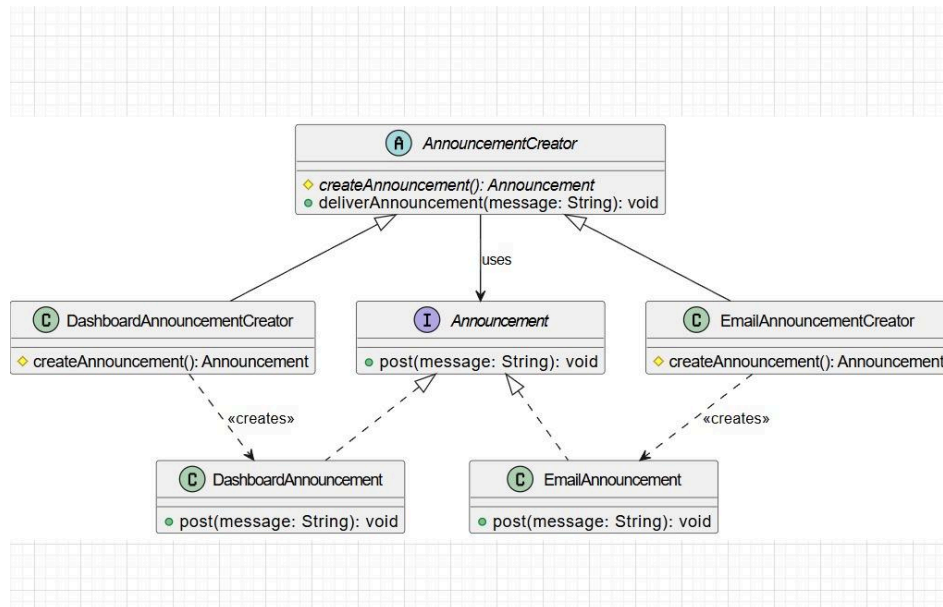
- Flexible creation
- Easy to extend
- Loose coupling

### Classes

- Announcement (interface)
- DashboardAnnouncement
- EmailAnnouncement
- AnnouncementCreator
- DashboardAnnouncementCreator

- EmailAnnouncementCreator

## UML Diagram



## Explanation

Creator decides which announcement to create.

## **5. Abstract Factory Pattern**

### **Case — Manage User Accounts**

#### **Problem**

System must create accounts for:

- Students
- Teachers

Each account needs:

- Profile
- Permissions

## **Solution**

Abstract Factory creates related objects together.

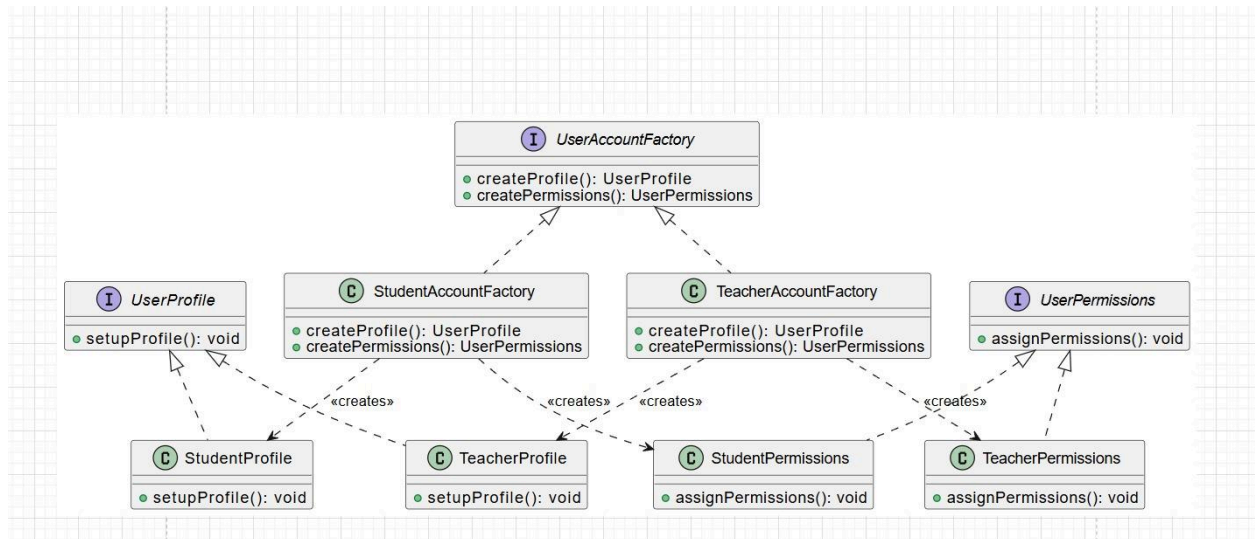
## **Why Abstract Factory?**

- Families of objects
- Easy to switch type
- Clean structure

## **Classes**

- UserProfile
- UserPermissions
- StudentProfile
- TeacherProfile
- StudentPermissions
- TeacherPermissions
- UserAccountFactory
- StudentAccountFactory
- TeacherAccountFactory

## **UML Diagram**



## Explanation

Factory creates profile + permissions together.

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## Case — Add Student Info UI

### Problem

System must work on:

- Web
- Mobile

Each UI needs:

- Form
- Validator

### Solution

Abstract Factory creates correct UI family.

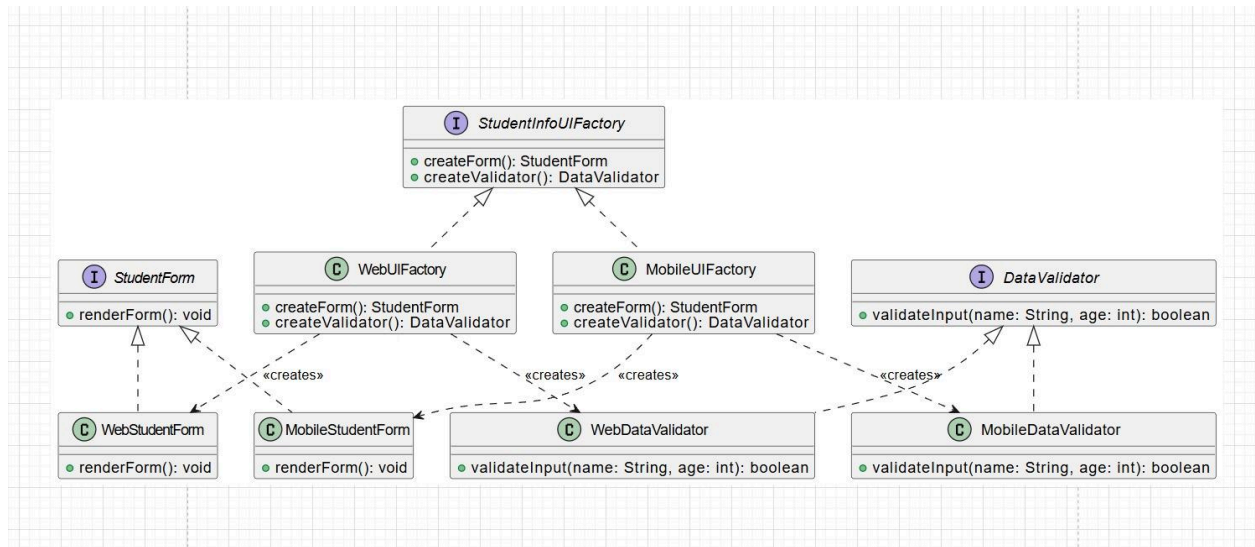
### **Why Abstract Factory?**

- Different platforms
- Same interface
- Easy extension

### **Classes**

- StudentForm
- DataValidator
- WebStudentForm
- MobileStudentForm
- WebDataValidator
- MobileDataValidator
- StudentInfoUIFactory
- WebUIFactory
- MobileUIFactory

### **UML Diagram**



## Explanation

Factory selects correct UI components.

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